Group 12

DRL Tracker
Vision Document

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Vision (Small Project)

1. Introduction

The purpose of this document is to collect, analyze, and define high-level needs and features of the DRL Tracker system. It focuses on the capabilities needed by the stakeholders and the target users, and why these needs exist. The details of how the DRL Tracker fulfills these needs are detailed in the use-case and supplementary specifications.

The introduction of the Vision document provides an overview of the entire document. It includes the purpose and references of this Vision document.

1.1 References

OU-SSupporter System.

- Developing Organization: Center for Information Systems Management, Mở University.
- Description: The primary reference system used for modeling core feature workflows.

HCMUS Student Portal.

- Developing Organization: VNUHCM - University of Science.
- Description: The existing institutional system used as the baseline for identifying current operational gaps, specifically its limitation to end of semester result viewing, which drives the necessity for the DRL Tracker solution.

University Training Point Evaluation Regulations.

- Issuing Body: VNUHCM - University of Science.
- Description: The official regulatory document outlining the criteria framework, scoring weights, and approval rules implemented within the DRL Tracker system.

2. Positioning

2.1 Problem Statement

The problem of	manual and fragmented management of university extracurricular activities and training points.
affects	students, activity organizers (Organizations), and assessment officers.
the impact of which is	easily lost proof of participation, difficult tracking of predicted points for students, and a time-consuming, error-prone point approval process.
a successful solution would be	a centralized tracking system (DRL Tracker) that automates point calculation, safely stores evidence, and streamlines the official point approval workflow.

2.2 Product Position Statement

For	university students and administration staff.
Who	need a reliable, transparent, and efficient way to manage training points.
The DRL Tracker	is a training point management system.
That	provides real-time predicted scores, centralized evidence storage, and an AI assistant for activity suggestions.
Unlike	traditional spreadsheet tracking, Google Forms, and physical paper certificates.
Our product	integrates the entire workflow from event registration and evidence submission to automated point calculation based on customizable university regulations.

3. Stakeholder and User Descriptions

3.1 Stakeholder Summary

Name	Description	Responsibilities
Admin	University IT or Administration staff.	Configures system rules, point regulations, and manages account permissions.
Assessment Officer - Faculty Level	Assessment staff responsible for reviewing student training point records within their assigned faculty.	Reviews Predicted Scores, evidence statuses, calculated points by criteria, and exceptional cases; proposes training point results; records reasons for adjustments requiring explanation; and forwards proposals to the university level.
Assessment Officer - University Level	Assessment staff responsible for reviewing faculty level proposals and finalizing student training point results at the university level.	Reviews proposals from faculties, cross checks supporting records, and approves and announces the final Official Score according to the applicable regulations.

3.2 User Summary

Name	Description	Responsibilities	Stakeholder
Student	End users who participate in university activities and track their training points.	Finds and registers for activities, submits participation evidence, tracks evidence status and Conduct score, and submits complaints or review requests when necessary.	Admin
Organization - Faculty Level	Event organizers responsible for activities within their assigned faculty.	Creates and manages activities, maintains participant and attendance lists, verifies student evidence against actual participation, provides rejection reasons, and handles initial complaints related to activities organized by the unit.	Assessment Officer - Faculty Level
Organization - University Level	Event organizers responsible for activities organized at the university level.	Creates and manages university level activities, maintains the common participant list for activities involving students from multiple faculties, verifies participation evidence, and handles initial complaints related to the unit's activities.	Assessment Officer - University Level
Assessment Officer - Faculty Level	Assessment staff responsible for reviewing student records within their assigned faculty.	Reviews predicted points and evidence, handles exceptional records within their authority, proposes training point results, records adjustment reasons when required, and forwards proposals to the university level.	Admin

Assessment Officer - University Level	Assessment staff responsible for university level assessment and final score approval.	Reviews faculty level proposals and supporting records, requests additional reviews when necessary, and finalizes the Official Score according to the applicable regulations.	Admin
Admin	Authorized personnel responsible for system administration.	Manages accounts and permissions, configures training point regulations and their versions, and maintains system level traceability.	University administration

3.3 User Environment

Students will access the system via web browsers on laptops or mobile devices. Organizations and Assessment Officers will primarily use desktop web browsers for management tasks.

3.4 Summary of Key Stakeholder or User Needs

- **Students:** need to track their training points in real time, distinguish Maximum Score, Predicted Score, and Official Score, receive notifications about activity schedules and evidence deadlines, and request reviews when evidence or calculated results are disputed.
- **Organizations:** need an efficient interface to manage activities, maintain accurate participant and attendance lists, verify evidence against actual participation, and handle initial complaints related to activities organized by their units.
- **Faculty level Assessment Officers:** need a centralized interface to review Predicted Scores, evidence statuses, calculated points by criteria, and exceptional records; propose training point results; document reasons for adjustments requiring explanation; and forward assessment records to the university level.
- **University level Assessment Officers:** need to review faculty level proposals and their supporting records, request additional reviews when necessary, incorporate relevant results from university level activities; approve and announce the final Official Score according to the applicable regulations.
- **Administrators (Admin):** need to manage accounts and permissions, configure applicable training point regulations and their versions, and ensure that important assessment actions and score adjustments are recorded for traceability and accountability.

Need	Priority	Concerns	Current Solution	Proposed Solutions
Track predicted points	High	Students cannot see their points before final approval.	Manual calculation by students.	Provide real-time predicted score based on regulations.
Verify participation evidence	High	Checking hundreds of paper proofs is slow and error-prone.	Physical paper certificates.	Centralized digital evidence storage and verification interface.
Approve official scores	High	Handling exceptions and syncing data is difficult.	Spreadsheets and manual data entry.	Automated point aggregation and one-click approval.

3.5 Alternatives and Competition

Current alternatives include manual tracking via Google Forms, Excel spreadsheets, and physical paper certificates. The major weaknesses of these alternatives include a lack of automated point calculation, easily lost paper proofs, and no real-time point tracking visibility for students.

4. Product Overview

The DRL (Conduct Point) Tracker system is a software platform designed to comprehensively support the management of students' training and extracurricular activities at universities and colleges. The system addresses issues related to storing participation evidence, automating the calculation of predicted Conduct scores, and standardizing the point verification and approval process according to the university's regulations.

The system serves four main user groups: Student, Organization, Assessment Officer, and Admin, ensuring transparency, accuracy, and traceability throughout the entire lifecycle of training points.

4.1 Product Perspective

DRL Tracker is an independent system but operates within the university's educational management ecosystem.

- **User Environment:** The system provides an interactive interface for students (prioritizing user-friendliness, easy lookup, and evidence submission) and a professional administration interface for organizations, assessment officers, and administrators.
- **System Interaction:** Although DRL Tracker can operate independently, it is assumed to require communication with the university's internal database (student information system) to synchronize student lists, student IDs, and faculty/class information.
- The system completely replaces current manual methods of managing evidence and calculating training points using paper documents or fragmented Excel files.

4.2 Assumptions and Dependencies

Assumptions:

- The university already has a clear set of training point regulations, with specific criteria and scoring, which the Admin can configure into the system.
- Users (Students, Staff) all have internet-connected devices (computers, smartphones) to access the system.
- Organizing units (Organizations) commit to using the system for taking attendance or issuing standardized evidence to students.

Dependencies:

- **Input Data:** The accuracy of the Predicted Score depends entirely on the honesty of the evidence submitted by the Student and accurate cross-checking by the Organization.
- **Personnel Process:** The system relies on the Assessment Officer performing timely reviews and approvals (Official Score) at the end of each semester to complete the point calculation cycle.

Additional Assumptions and Dependencies:

- **Applicable Training Point Regulations:** The system assumes that the training point regulations and assessment criteria applicable to each semester are provided and configured by authorized personnel. The system must apply the correct regulation version when calculating and reviewing training point results.
- **Historical Assessment Records:** Historical Official Scores must not be automatically overwritten or recalculated using a newer regulation version.
- **Assessment Workflow Dependency:** The finalization of student training point results depends on the review and approval workflow between faculty level and university level Assessment Officers. Faculty level proposals must be forwarded with their supporting records, while the university level Assessment Officer is responsible for final approval and announcement of the Official Score.

- **Supporting Evidence and Activity Records:** The accuracy of assessment results depends on the reliability of activity participant lists, attendance records, and supporting evidence provided or verified by the relevant Organizations and students. The system supports verification and traceability but cannot independently guarantee the truthfulness of externally provided information.

5. Product Features

Below are the main features (High-level capabilities) of the DRL Tracker system, categorized by user groups:

Feature Group for Students

- **FEA-01. Activity Discovery and Registration:** Allows students to search, view details, and register/unregister for extracurricular activities by time, organizing unit, or criteria.
- **FEA-02. Evidence Management:** Ability to upload image/document evidence for activities; track evidence status (Pending, Valid, Invalid); submit additional evidence if necessary.
- **FEA-03. Training Point Tracking:** View aggregated points by semester/academic year/criteria. The system clearly distinguishes 3 types of scores: Maximum Score, Predicted Score, and Official Score.
- **FEA-04. Complaints and Feedback:** Submit review requests (up to 5 times) for evidence assessed as invalid or when there are score discrepancies.
- **FEA-05. AI Assistant:** A chatbot assisting in consulting and recommending suitable activities based on the student's preferences, schedule, and missing training point criteria in the current semester.

Feature Group for Organizations

- **FEA-06. Activity Lifecycle Management:** Create new activities, configure information (time, capacity, criteria), submit for approval, edit, or cancel activities.
- **FEA-07. Participant List Management:** Track the list of registered students, update the actual participant list (attendance), and finalize the list after the event concludes.
- **FEA-08. Evidence Verification:** Cross-check students' evidence against the actual participant list, update the status to valid/invalid, along with rejection reasons.
- **FEA-09. Complaint Handling & Profile Forwarding:** Enables Organizations (Faculty/University level) to verify evidence and resolve student complaints for their managed activities (university activities are unified, not duplicated for each faculty). Organizations can request extra evidence, update decisions with reasons.

Feature Group for Assessment Officers

- **FEA-10. Activity Moderation:** Review and approve activities proposed by Organizations before they are published to students.
- **FEA-11. Exceptional Record Processing:** Review, verify, and award points for evidence belonging to activities outside the system (Olympic competitions, off-site blood donation, academic achievements, etc.).
- **FEA-12. Point Aggregation and Proposal (Faculty Level):** Enables Faculty Assessment Officers to review students' predicted scores, evidence statuses, and exceptional cases within scope. Officers can verify records, resolve discrepancies within authority, propose final scores with adjustment reasons, and forward proposals to University Assessment Officers for final approval.
- **FEA-13. Official Score Approval (University Level):** Enables University Assessment Officers to review faculty proposals, cross check evidence, request re-reviews, and integrate university level activity results. They approve and announce final Official Scores per regulations. Any score adjustments require documented reasons, and full processing history must be retained for traceability and accountability.

Feature Group for Administrators (Admin)

- **FEA-14. Regulation Configuration Management:** Create and manage versions of training point regulations (criteria, weights, addition/subtraction rules, maximum score limits); assign regulation versions to each semester. Historical Official Scores must not be automatically overwritten or recalculated using a newer regulation version.

- **FEA-15. Account & Authorization Management:** Create, verify, and grant permissions to sensitive accounts (Organization, Assessment Officer).
- **FEA-16. System History Management (Audit Trail):** Record and track the operational history of accounts (especially point approval/adjustment decisions) for traceability and accountability.

6. Non-Functional Requirements

Security & Privacy Requirements

- **Personal Data Protection:** Image evidence (which may contain personal information) must have restricted access. Only the owning Student, the relevant Organization, and the Assessment Officer are allowed to view it.
- **Audit Trail:** All critical actions such as granting permissions, changing regulations, rejecting evidence, approving or adjusting the Official Score must be logged by the system (Who did it? When? What is the reason for the adjustment?).
- **Data Integrity:** Training point data and regulations from previous versions (semesters/academic years) must absolutely not be overwritten or automatically recalculated when a new regulation is issued.
- **Strict Authorization:** Assessment Officers and Admins must be strictly verified. The AI Assistant only has "read/consult" access, with no rights to write, edit, or make decisions regarding scores and evidence. The Admin does not have the right to directly edit the approved Official Score.

Performance Requirements

- The system must ensure good load-bearing capacity and fast response times during peak periods (e.g., at the beginning of the semester when students mass-register for activities, or at the end of the semester when students submit evidence, check scores, and file complaints).

Usability Requirements

- **User Interface:** It is necessary to clearly distinguish the 3 types of scores (Maximum Score, Predicted Score, Official Score) on the Student's interface by using annotations, colors, or visual charts to avoid confusion.
- **Automated Alerts:** The system must be capable of sending real-time or periodic notifications/reminders (e.g., reminding of evidence submission deadlines, upcoming activities, score update notifications).

Retention & Regulatory Requirements

- Evidence data and training point records must be stored for at least the duration of the student's study cycle at the university (typically 4-5 years) to serve inspection and graduation review purposes.